RAG workshop
Presenter guide

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How the workshop works

Use the slides to explain one idea, then switch to the application for the action. The guide contains the spoken script and exact demonstration steps. Installation and deployment happen before the audience arrives. Present in English. Keep the source PDF and live document questions in French.

Session timing

Slides 1-9: orientation, cloning and project tour, 17 minutes.
Slides 10-13: document, chunks, vectors and storage, 13 minutes.
Slide 14: preparation in the application, 5 minutes.
Slides 15-20: retrieval, answers, verification and experiment, 17 minutes.
Slide 21: questions and discussion, 8 minutes.

Before the session

1. Clone the GitHub repository, follow the setup page and configure .env privately.
2. Confirm the PDF and working Python environment.
3. Prepare the cache, start Streamlit and test one real question shortly before the session.
4. Leave PowerPoint, the project folder, the PDF and the browser ready.
5. Keep Recompute embeddings unchecked during the demonstration.

When time is short

Keep the first question, source check and missing-fact test. Make the three-versus-five experiment optional. If an endpoint is unavailable, explicitly use the recorded application views on slides 14 and 17 and the source quotation on slide 18.

Presenter View

Each slide has the same speaking notes embedded in PowerPoint. The following slide pages add expected results and fallback instructions where useful.

Demo run card

1. Read and inspect - slide 10

Open A-6.002.pdf. Show that text can be selected. In Streamlit choose the same document and confirm 18 pages with text.

2. Try chunking - slide 11

Open Settings. Keep overlap at 15. Change Words per passage from 80 to 120. The count changes from 110 to 72. Restore 80. This is a local preview, so it does not make an embedding call. Changing chunk settings invalidates the active assistant, which is why this exercise comes before preparation.

3. Prepare - slide 14

Keep Passages per answer at 3. Click Start processing once. Wait for Ready and 110 passages. Cached embeddings may be reused. A first run or an idle endpoint can take longer. Keep the same browser tab after preparation.

4. Ask and verify - slides 17-18

À quelle fréquence le personnel renouvelle-t-il ses engagements de confidentialité ?

Expected fact: chaque année. Expand View retrieved passages. Find page 5 and compare the original PDF. Technical source ID for the presenter: a-6_002_p0005_0195.

5. Test a missing fact - slide 19

Click New conversation. Ask: Quel est le mot de passe du Wi-Fi de Revenu Québec ?

Expected: Je ne trouve pas cette information dans les documents.

6. Compare retrieval depth - slide 20

Repeat the same annual-renewal question in a new conversation with 3, then 5 passages per answer. Compare the evidence and the claim. Restore 3. Do not rebuild embeddings. No improvement is guaranteed.

Clone and set up the project

Before the workshop

Install Git and Python 3.12 or 3.13. Open PowerShell in the parent folder where you want to keep the project. On macOS or Linux, use the platform-specific commands in README.md.

```
git clone `
  https://github.com/Abdourahmandev/RAG_assistant_workshop.git
cd RAG_assistant_workshop
py -3.13 -m venv .venv
$py = ".\.venv\Scripts\python.exe"
& $py -m pip install -r requirements.txt
& $py -m pip check
Copy-Item .env.example .env
```

Use py -3.12 if that is your installed version. The backtick continues the clone command on the next line. For an existing clone, skip cloning and copying .env again.

Private configuration

Open .env privately and enter the authorized values supplied by the instructor. Use deployment root URLs. The client adds the API routes.

```
HF_CHAT_ENDPOINT=https://<chat>.endpoints.huggingface.cloud
HF_EMBED_ENDPOINT=https://<embed>.endpoints.huggingface.cloud
HF_TOKEN=<your-authorized-token>
```

Inspect, prepare and launch

```
& $py app.py --inspect
& $py app.py --prepare-only
& $py -m streamlit run streamlit_app.py
```

Open http://127.0.0.1:8501 and click Start processing. The first preparation creates the local embedding cache and database. Streamlit reuses the embedding cache and prepares its own index. Allow time for the initial run before the session. If Streamlit is already running, reuse its browser tab.

Troubleshooting and wrap-up

Cold start or interrupted preparation

An idle Hugging Face endpoint may need several minutes to start. The application retries transient service errors. Completed embedding batches remain cached. If preparation fails, inspect the displayed error and service state, then retry once the cause is resolved. Do not enable Recompute embeddings merely because a batch was interrupted.

Authentication or model errors

Check the token permissions and deployment root URLs privately. Confirm Qwen/Qwen3-1.7B and google/embeddinggemma-300m in the Hugging Face deployment configuration. The application adds the API routes. The key is HF_EMBED_ENDPOINT, including the final T.

If the answer is wrong

Inspect retrieved passages first. Missing evidence points toward extraction, chunking or retrieval. If the right evidence is present but the answer misstates it, inspect the generation instructions and model behavior. Citations alone do not establish correctness.

After the workshop

Stop Streamlit with Ctrl+C in its terminal when finished. Manage the hosted endpoints separately. Stopping the local application does not pause them. Keep tokens out of slides and shared project copies.

SLIDE 01 / 21 - 1 MIN

Build a RAG Assistant

Say

Today we will turn a PDF into an assistant that can answer questions and show its evidence. We will follow one document through the whole process. For each stage, I will explain what it does, why we chose this approach, and what to look for in the application. By the end, you should be able to explain where an answer came from and identify which part of the process to improve.

Do

Show the cover. Keep the application and PDF ready in separate windows. Present in English. Keep the document questions and answers in French.

Transition

Here is how we will work together.

SLIDE 02 / 21 - 1 MIN

Abdourahman H. Mawlid

Say

I am Abdourahman H. Mawlid, Senior Data Engineer and founder of Decidata. This is a guided workshop. We will alternate between a short explanation and an action in the project. The environment and model endpoints should already be configured. We will spend about fifty-two minutes exploring and practicing, then leave eight minutes for your questions.

Do

Introduce yourself briefly. Explain that participants will use Streamlit and the repository, with no notebook required.

Transition

Let us begin with the kind of information we want to query.

SLIDE 03 / 21 - 1 MIN

Traditional databases

Say

When information already has a clear structure, a database query is a natural tool. For example, an orders table can tell us how many orders shipped today. Our example today is a privacy policy written in paragraphs. The answer is in the document, but we first need to find the right passage. That is the problem retrieval helps us solve.

Do

Use the orders example briefly. Ask participants to think of a document they regularly search.

Transition

A language model can explain a passage once we give it the relevant text.

SLIDE 04 / 21 - 2 MIN

Why an LLM needs retrieval

Say

RAG means retrieval-augmented generation. Retrieval finds useful evidence in our documents. Generation uses that evidence to compose an answer. The application sends the question and selected passages to the model. This lets us work with a document without training the model on it. The answer still needs checking because either the search or the generation can make a mistake.

Do

Follow the existing diagram. Ask: where does the information from our PDF enter the process?

Transition

The full workflow has a preparation path and a question path.

SLIDE 05 / 21 - 3 MIN

The RAG pipeline

Say

The top row prepares the document. We read and split it, turn each passage into a vector, and import the passages, vectors and page references into a vector database. That final import is what indexing means in this workshop. A vector is a list of numbers that helps us compare texts by meaning. The bottom row runs for each question. We embed the question, search the stored vectors, and send the retrieved text to the language model. The document vectors can be reused across many questions.

Do

Point to the vector database. Distinguish storage from generation. Ask participants which row runs once for a prepared document and which row runs for every new question.

Transition

Each stage has several possible strategies. Here are the choices we will use today.

SLIDE 06 / 21 - 3 MIN

Choices at each RAG step

Say

There is no single best configuration for every document. A scan needs OCR, while our PDF already contains text. Chunks can follow a fixed length, paragraphs or document sections. Embeddings should suit the language and subject. Storage can be local or shared. Retrieval can use meaning, exact words, or a combination. Generation can write a cited answer, extract a quotation, or return a fixed structure. We will use the middle column and explain the reason as we reach each stage. These are options we can combine, not exclusive categories.

Do

Spend no more than three minutes on this map. Point to the chosen column. Do not explain every alternative in depth yet.

Transition

First, let us get the same project onto each computer.

Reference

https://pypdf.readthedocs.io/en/stable/user/extract-text.html

https://huggingface.co/google/embeddinggemma-300m

https://docs.trychroma.com/docs/overview/introduction

SLIDE 07 / 21 - 2 MIN

Getting the project from GitHub

Say

The workshop project is available on GitHub. Before the session, install Git and Python 3.12 or 3.13. Open PowerShell in the folder where you keep your projects. Run git clone with this repository URL, then change into RAG_assistant_workshop. Cloning downloads the code, sample PDF and presentation materials. Follow the README to create the virtual environment and install the dependencies. Copy .env.example to .env and enter your authorized Hugging Face endpoint URLs and token privately. Then start Streamlit using the command in the README. The first processing run creates the embeddings and local database on your computer, so allow time for preparation before the live exercises.

Do

Show the GitHub repository. In PowerShell, demonstrate the clone and cd commands, or show the already cloned folder. Open README.md and point to Windows PowerShell setup. Complete installation before the session. Keep .env closed when presenting. Start Streamlit after configuration.

Look for

The cloned folder contains app.py, streamlit_app.py, rag/, tests/, the PDF and the workshop materials. Each participant configures .env privately and prepares their own index.

Transition

Now let us locate each RAG responsibility in the cloned project.

Reference

https://github.com/Abdourahmandev/RAG_assistant_workshop

SLIDE 08 / 21 - 2 MIN

The repository at a glance

Say

The source PDF is our input. streamlit_app.py is the interface we will use during the workshop. app.py is a terminal alternative. The rag folder contains the shared pipeline, so both interfaces use the same logic. Inside it, corpus reads the document, chunking splits it, pipeline coordinates embedding and retrieval, and vector_store handles storage and search. The local database folder appears after preparation. You only need to recognize these responsibilities today, not read every function.

Do

Open the project folder in the editor. Expand rag/. Point to the PDF, streamlit_app.py and the pipeline modules. Keep .env closed. Then switch to the running Streamlit application.

Transition

Some parts run on this laptop, while the models run on Hugging Face.

SLIDE 09 / 21 - 2 MIN

Two models hosted on Hugging Face

Say

Both models in this project run on Hugging Face Inference Endpoints. EmbeddingGemma turns document passages and questions into vectors. Qwen writes the final answer using retrieved text. The PDF reader, Streamlit interface and Chroma database run on the laptop. This setup avoids requiring a local GPU for the workshop. A local deployment is another choice when hardware or data constraints call for it. Today, inference requires an internet connection and sends the necessary text to our configured endpoints.

Do

Point to each model and ask which one finds related text and which one writes. If showing the Hugging Face dashboard, show model names and status only. Do not show credentials or spend workshop time creating deployments.

Transition

Our first practical step is to look at the document itself.

Reference

https://huggingface.co/google/embeddinggemma-300m

https://huggingface.co/Qwen/Qwen3-1.7B

https://huggingface.co/docs/inference-endpoints/index

SLIDE 10 / 21 - 3 MIN

1. Reading the source document

Our strategy and reason

Text extraction and light cleaning. The PDF already contains selectable text. Preserve page references so each passage remains traceable.

Say

Our source is an eighteen-page privacy policy from Revenu Québec. Reading the PDF means extracting its text while retaining the page it came from. We normalize whitespace and keep the original page references. This PDF has selectable text, so we can use a text extractor. A scanned page would require OCR. Extraction can still damage punctuation or reading order, so checking the original matters. Keeping page references will let us verify the answer later.

Do

Open A-6.002.pdf and select a sentence to show that it contains text. In Streamlit select the same file. Confirm Pages with text is 18. Do not start processing yet.

Look for

The application recognizes 18 pages with text.

Transition

Next we divide that text into passages the retriever can search.

Reference

https://pypdf.readthedocs.io/en/stable/user/extract-text.html

SLIDE 11 / 21 - 4 MIN

2. Splitting text into chunks

Our strategy and reason

Fixed-size sliding windows: 80 words, 15-word overlap, within one page. This creates simple, repeatable boundaries and retains some context near each boundary. Compare with paragraph, section or semantic chunking.

Say

A chunk is a short passage we can retrieve independently. We use windows of eighty words. The next window repeats fifteen words from the previous one, which helps retain context near a boundary. Smaller chunks can isolate a fact. Larger chunks can preserve more context but may mix topics. Paragraphs and sections are useful alternatives when the document structure is reliable. Our fixed windows are easy to inspect and give us a simple starting point for this workshop.

Do

Expand Settings in Streamlit. Keep Overlap in words at 15. Change Words per passage from 80 to 120 and observe the passage count change from 110 to 72. Restore 80 before continuing. This preview is local and does not call the embedding endpoint. Do not click Recompute embeddings.

Look for

80 words gives 110 passages. 120 words gives 72. Restore 80 words and 15 overlap.

Transition

The search engine now needs a numerical representation of every passage.

SLIDE 12 / 21 - 3 MIN

3. Turning text into vectors

Our strategy and reason

Dense multilingual embeddings with EmbeddingGemma 300M, using 768 dimensions. Use the same model for passages and questions, with their task-specific prefixes. This suits the French corpus and meaning-based retrieval. Embed in batches of 8 and cache completed batches to avoid repeating work. Compare with an English-only or domain-tuned embedding model.

Say

An embedding is a list of numbers representing text for comparison. Our EmbeddingGemma deployment produces seven hundred and sixty-eight numbers per passage. These displayed values are the first four numbers from the saved vector for the page-five passage. Individual numbers are not readable facts. The model uses the whole vector to help rank related text. We use the same model for the document and the question, with the appropriate input formatting. A multilingual model suits our French document. A specialized model may help with a different domain, but it would need evaluation.

Do

Point to the real vector excerpt on the slide. Ask: which model should also embed the question? Explain that the next preparation action will create or reuse these vectors and store them.

Transition

A useful search record keeps the text and its origin alongside that vector.

Reference

https://huggingface.co/google/embeddinggemma-300m

SLIDE 13 / 21 - 3 MIN

4. Importing into a vector database

Our strategy and reason

Persistent local Chroma storage for text, vectors and source metadata. The implementation uses an HNSW index with cosine distance. Local storage is convenient for a single workshop laptop and allows reuse. Compare with a managed vector store or SQL database with vector support.

Say

Indexing imports the prepared records into Chroma. Each record contains the passage text, its embedding and metadata such as the PDF page. The vector helps us find related passages. The text gives the language model evidence to read. The page tells us where to check it. We choose a local persistent database because it is convenient for one laptop and one workshop document. A shared application may need a managed service or a database with vector support. The database stores evidence. The language model still writes the answer.

Do

Point to all three fields. Open the base_vectorielle folder only if useful, without opening or editing the database files. Explain that Start processing completes embedding and database import together.

Transition

We can now run the preparation path in the application.

Reference

https://docs.trychroma.com/docs/overview/introduction

SLIDE 14 / 21 - 5 MIN

Practice: preparing the document

Say

Now we run the top row of the pipeline. Check that the document has eighteen pages and the current settings give one hundred and ten passages. Click Start processing once. The application checks the models, creates missing embeddings, and imports the records into its local database. This laptop already has completed embeddings, so it can reuse them. A new installation will take longer, and an idle hosted model may need time to start. When the assistant says Ready, preparation is complete.

Do

In Streamlit confirm 80 words, overlap 15 and Passages per answer 3. Keep Recompute embeddings unchecked. Click Start processing once. Wait for Ready and 110 passages. If already prepared, explain that the saved preparation is being reused. Do not launch a second server.

Look for

Assistant: Ready. 110 passages available. A prepared laptop may report 110 reused from cache.

If the live step is unavailable

If the endpoints cannot wake during the session, show the recorded Ready view. Explain that it is a saved demonstration, then continue with the recorded answer and source examples.

Transition

From now on, each question follows the retrieval and answer path.

SLIDE 15 / 21 - 3 MIN

5. Retrieving useful passages

Our strategy and reason

Dense semantic retrieval using cosine similarity, with top k = 3. This keeps the evidence short enough to inspect. The implementation has no reranker or confidence threshold. Compare with keyword retrieval, hybrid search, reranking or a different k.

Say

For a question, we first create its vector and search the stored document vectors. Top three means we return the three highest-ranked passages. It does not mean all three are equally useful or that any answer is guaranteed. These page descriptions come from a recorded run of our French question. Page five contains the annual renewal statement. Page four discusses confidentiality laws. Page seventeen discusses destruction and policy changes. Which one actually answers how often staff renew their commitments? Meaning-based retrieval is a simple starting point. Exact keywords, hybrid search and reranking are alternatives when retrieval misses evidence.

Do

Read the question: À quelle fréquence le personnel renouvelle-t-il ses engagements de confidentialité ? Ask the audience to choose the supporting page before revealing the answer in the application. Accept page 5. Explain that rankings can change with wording or settings.

Transition

The generator receives the selected text and the question together.

SLIDE 16 / 21 - 2 MIN

6. Writing an answer from evidence

Our strategy and reason

Evidence-grounded prompting with Qwen3 1.7B: a concise French answer, source citations and abstention when evidence is absent. The application uses temperature 0 and disables thinking mode. These choices support a predictable workshop flow, but do not guarantee correctness. Compare with extractive answers or structured output.

Say

Qwen receives the question, the retrieved passages and instructions. We ask it to answer in French, use only supported information, cite its sources and abstain when evidence is missing. The model does not search the whole PDF at this stage. It sees the context supplied by retrieval. We choose a short cited answer because the audience needs to inspect the source. For other tasks, we might ask for a direct quotation or a structured response. Prompt instructions help, but we still need to verify the output.

Do

Explain the three inputs in plain language. Ask what happens if retrieval does not include the passage with the answer.

Transition

Let us ask the question and inspect the result.

Reference

https://huggingface.co/Qwen/Qwen3-1.7B

SLIDE 17 / 21 - 4 MIN

Practice: our first question

Say

Let us ask our prepared assistant how often the staff renew their confidentiality commitments. The French wording is shown here. The expected fact is every year. The application first retrieves passages and then generates the answer. The screenshot records a real exchange from this project. During the live demonstration, your wording may differ, but the factual claim should be supported by the same source. Do not judge the answer only by how fluent it sounds.

Do

Click New conversation if a previous exchange is present. Ask: À quelle fréquence le personnel renouvelle-t-il ses engagements de confidentialité ? Wait for the answer. Look for chaque année and a source citation.

Look for

An answer in French stating that renewal happens every year, with a citation to the page-five passage.

If the live step is unavailable

If the network is unavailable, state that the displayed screenshot is a recorded response. Continue to the source-checking exercise.

Transition

A citation gives us a place to check. We still need to read it.

SLIDE 18 / 21 - 3 MIN

7. Checking the answer

Our strategy and reason

Human source verification plus an unsupported-question test. Check that the passage supports the claim and that the assistant abstains when appropriate. This makes the demonstration inspectable. A production evaluation would need more documents, questions and reviewers.

Say

Page five says that staff sign confidentiality commitments and renew them every year. We compare that source statement with the answer. First, does the cited passage exist? Second, does it support the specific claim? The application displays other retrieved passages too, so find the one actually cited. This check distinguishes a convincing answer from an answer we can justify. The source remains in French so that the live claim and its evidence can be compared directly.

Do

Expand View retrieved passages. Find A-6.002.pdf, page 5. Read chaque année. Open page 5 of the original PDF and compare the statement. Ask a participant to explain why the citation supports the answer.

Look for

The PDF and the answer agree on annual renewal.

Transition

A useful assistant also needs to recognize when the document cannot answer.

SLIDE 19 / 21 - 2 MIN

Practice: a missing fact

Say

We now ask for something the policy does not contain: the Wi-Fi password. Start a new conversation so the example is easy to interpret. The expected response says that the information is not in the documents. This is the right behavior for this question. However, the same refusal can also happen when retrieval fails to find a fact that is present. That is why we test both answerable questions and deliberately unsupported ones.

Do

Click New conversation. Ask: Quel est le mot de passe du Wi-Fi de Revenu Québec ? Verify the French abstention. Confirm that the assistant remains Ready.

Look for

Je ne trouve pas cette information dans les documents.

Transition

We can also test a retrieval choice without preparing the PDF again.

SLIDE 20 / 21 - 3 MIN

Practice: three passages or five?

Say

Retrieval depth controls how much evidence reaches the model. Three passages keeps the context short. Five may add a useful passage, but it can also add irrelevant material. We will change only that setting and repeat the same question. We are looking at the retrieved evidence as well as the wording of the answer. More context does not automatically mean a better answer. Changing the number of retrieved passages reuses the existing document vectors.

Do

Start a new conversation and ask the annual-renewal question with Passages per answer set to 3. Inspect the sources. Change it to 5, start a new conversation and repeat the exact question. Compare the sources and supported claim. Restore 3. Do not click Recompute embeddings. Allow this as an optional exercise if startup took too long.

Look for

The panel shows up to five retrieved passages. No specific improvement is promised. The source-supported fact should stay the same.

Transition

You now have a method for trying a change and checking its effect.

SLIDE 21 / 21 - 8 MIN

Your document, your questions

Say

We have followed the complete RAG workflow: read a document, split it, create vectors, import them into a database, retrieve passages and generate an answer. The final step is checking whether the evidence supports the claim. For your own document, begin with one question you can verify and one question the document cannot answer. When a result is poor, first inspect the retrieved passages. That helps you decide whether to change extraction, chunking, retrieval or the answer instructions. What would you try with your own documents?

Do

Invite questions and discussion. Ask participants to name one setting they would change and how they would judge the result. Keep commands and troubleshooting in the separate presenter guide.

Transition

End the session with the project and guide available to participants.